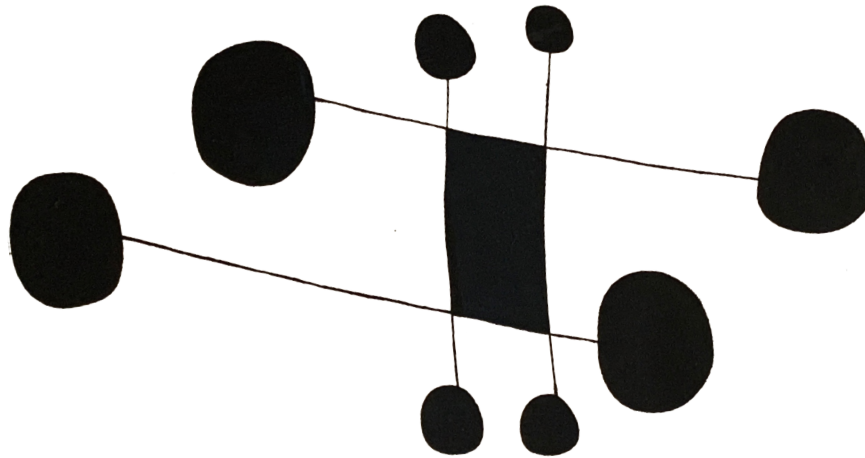


Generalized Kähler Metrics and Double Symplectic Groupoids



Noncommutative Shapes
Antwerp, Sep 13, 2022

Marco Gualtieri
University of Toronto

Generalized Complex Structures

M^{2n} smooth manifold $(\pi M = TM \oplus T^*M, \langle, \rangle)$ $O(2n, 2n)$

$\mathbb{J} : \pi M \rightarrow \pi M$ $\mathbb{J}^2 = -1$ Courant integrable

$\downarrow \mathbb{J}$
 $U(n, n)$

e.g.: $\mathbb{J} = \begin{pmatrix} I & 0 \\ 0 & -I^* \end{pmatrix}$ I complex str.

Deformations: $T_{[I]} \mathcal{M}_{GC} = H^0(\wedge^2 \tau) \oplus H^1(\tau) \oplus H^2(\mathcal{O})$

ON GLOBAL DEFORMATION QUANTIZATION IN THE ALGEBRAIC CASE

MICHEL VAN DEN BERGH

ABSTRACT. We give a proof of Yekutieli's global algebraic deformation quantization result which does not rely on the choice of local sections of the bundle of affine coordinate systems. Instead we use an argument inspired by algebraic De Rham cohomology.

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Integration via decorated surfaces à la Ševera *

Double Morita equivalence with complex conjugation *

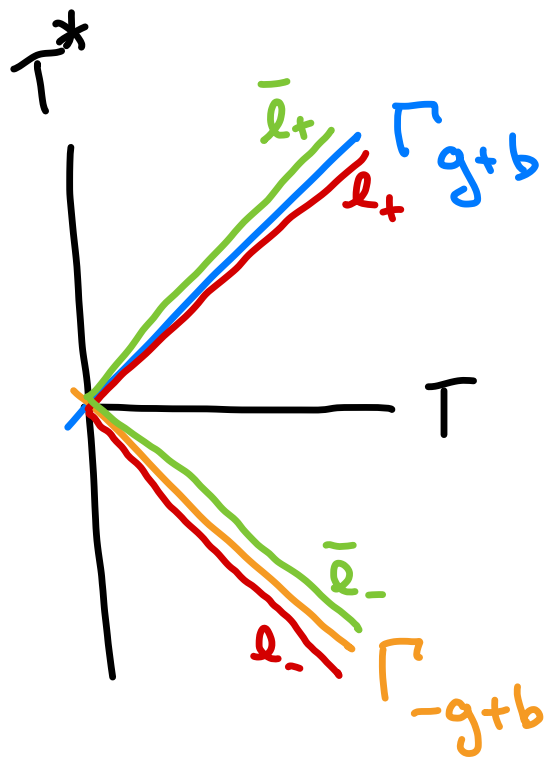
Double magnetic deformation: The 2D Bursztyn shuffle *

Generalized Kähler structures

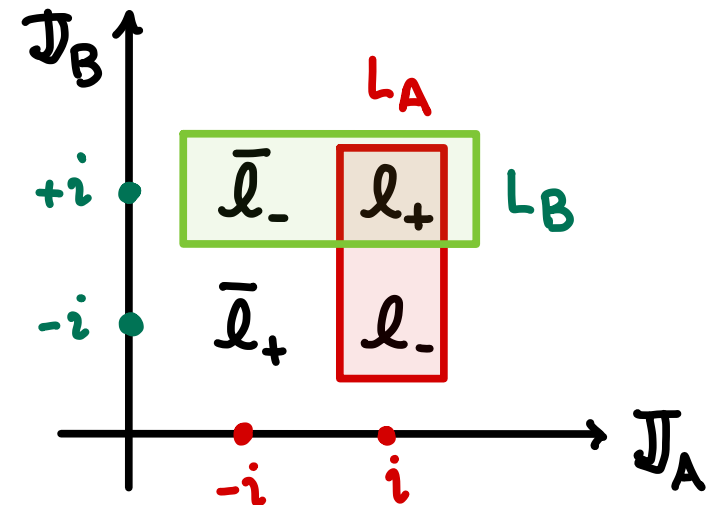
- (M, H) C^∞ mfd, $H \in \Omega^3(M)$ $dH=0$
 $(\pi M = TM \oplus T^*M, \langle, \rangle, [,]_H)$ Courant algebroid
 - $(\mathbb{J}_A, \mathbb{J}_B)$ pair of generalized complex structures
s.t.
$$\begin{cases} \mathbb{J}_A \mathbb{J}_B = \mathbb{J}_B \mathbb{J}_A \\ \langle \mathbb{J}_A^-, \mathbb{J}_B^- \rangle > 0 \end{cases}$$
 - E.g.: $\mathbb{J}_A = \begin{pmatrix} 0 & -\omega^{-1} \\ \omega & 0 \end{pmatrix}$ (ω, I) Kähler $g = -\omega I$
 $\mathbb{J}_B = \begin{pmatrix} I & 0 \\ 0 & -I^* \end{pmatrix}$
- $$\langle \mathbb{J}_A(X + \xi), \mathbb{J}_B(Y + \eta) \rangle = \frac{1}{2} (g(X, Y) + g^{-1}(\xi, \eta))$$

Bihermitian Geometry (Gates—Hull—Roček 1984)

$$((M, H), (g, b, I_+, I_-))$$



g Riemannian
 $b \in \Omega^2(M, \mathbb{R})$

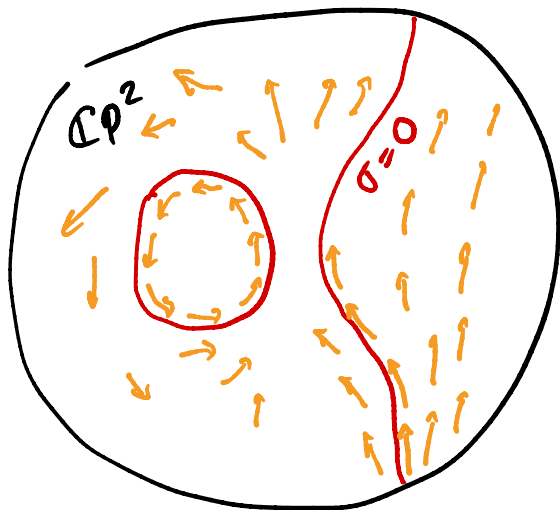


$$\begin{array}{cc} l_+ \oplus \bar{l}_+ & l_- \oplus \bar{l}_- \\ \downarrow & \downarrow \\ (I_+ & , & I_-) \end{array}$$

Pair of g -orthog.
 complex str.

$$g I_{\pm} = \omega_{\pm} \quad \text{satisfy} \quad \pm d^c \omega_{\pm} = H$$

Example 1: GK structure on $\mathbb{CP}^2$



Initial data:
$$\begin{cases} (M, I_-) = \mathbb{CP}^2 \\ \sigma \in H^0(\wedge^2 T) = H^0(\mathbb{P}^2, \mathcal{O}(3)) \\ \omega_0 \text{ Fubini-Study Kähler structure} \end{cases}$$

$$[\sigma(\omega_0)] \in H^1(T) = 0$$

$$\Rightarrow \exists v \in C^\infty(TM) \text{ s.t. } \begin{cases} \bar{\partial} v = \sigma(\omega_0) \\ [v, \sigma] = 0 \end{cases}$$

Output data:
$$\left\{ \begin{aligned} I_+ &= \varphi_1^v(I_-), & g &= -\bar{\omega} \left(\frac{I_+ + I_-}{2} \right), & \bar{\omega} &= \int_0^1 (\varphi_s^v)^* \omega_0 \, ds \end{aligned} \right\}$$

$$(g, I_+, I_-) \text{ Generalized Kähler on } \mathbb{CP}^2$$

Modules over regular algebras of dimension 3

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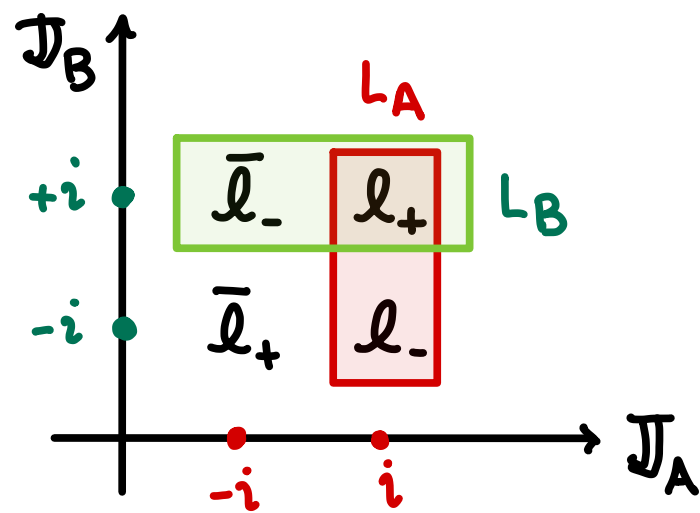
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1 Introduction

Let k be a field. In a previous paper [ATV] (see also [OF]) some graded k -algebras A , regular algebras of dimension 3, were constructed from certain automorphisms σ of elliptic curves or of more general one-dimensional schemes E with arithmetic genus 1, which are embedded as cubics in $\mathbb{P}^2$ or as divisors of bidegree $(2, 2)$ in $\mathbb{P}^1 \times \mathbb{P}^1$. In this correspondence, the points of the scheme E were shown to parametrize certain A -modules called *point modules*. A point module N is a graded right A -module with these properties:

Holomorphic reduction and Manin triples



$\mathfrak{l}_+ \cong T_{0,1}(\mathcal{I}_+)$ acts on πM

$\Rightarrow$ Courant reduction

"Isotropic extended action"

reduction data: $K = \mathfrak{l}_+$

$$E_{\text{red}} = \left(\mathfrak{k}^\perp / \mathfrak{k} \right)^K =$$

$$\left(\begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \square \\ \hline \end{array} \bigg/ \begin{array}{|c|} \hline \square \\ \hline \end{array} \right)^K$$

$\mathcal{I}_+$ - hol. sections

of $\mathfrak{k}^\perp / \mathfrak{k}$

$$(\mathfrak{L}_A)_{\text{red}} = \left(\begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \square \\ \hline \end{array} \bigg/ \begin{array}{|c|} \hline \square \\ \hline \end{array} \right)^K$$

$$(\mathfrak{L}_B)_{\text{red}} = \left(\begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \square \\ \hline \end{array} \bigg/ \begin{array}{|c|} \hline \square \\ \hline \end{array} \right)^K$$

Holomorphic reduction and Manin triples

$$\mathcal{E}_+ = \mathcal{A}_+ \oplus \mathcal{B}_+$$

$$\downarrow$$
$$X_+$$

$$\mathcal{E}_- = \mathcal{A}_- \oplus \mathcal{B}_-$$

$$\downarrow$$
$$X_-$$

Result of Courant reduction (G , following B.-C.-G.)

Complex manifolds $X_{\pm}$

Holomorphic Manin triples

$$\mathcal{E}_{\pm} = \mathcal{A}_{\pm} \oplus \mathcal{B}_{\pm}$$

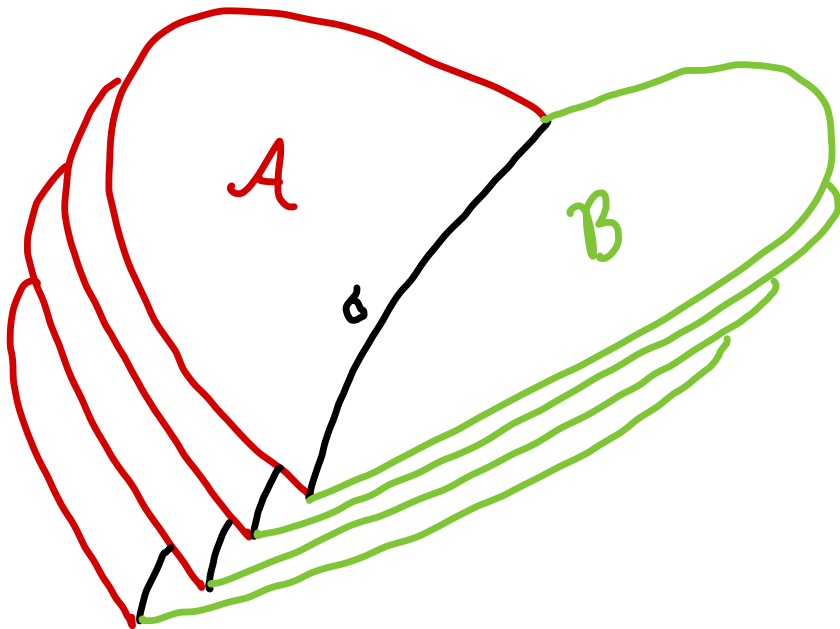
A pair of transverse holomorphic singular foliations

$$A_{\pm}, B_{\pm}$$

induce transverse singular foliations on $X_{\pm}$

$$A_{\pm} - B_{\pm} = \Gamma_{G_{\pm}}$$

Intersecting in the symplectic foliation of the Hitchin Poisson structures



Main Questions:

$$\mathcal{E}_+ = \mathcal{A}_+ \oplus \mathcal{B}_+$$

$\downarrow$

$$X_+$$

$$\mathcal{E}_- = \mathcal{A}_- \oplus \mathcal{B}_-$$

$\downarrow$

$$X_-$$

1. How are these holomorphic Manin triples related?
2. What is the role of the (non-holomorphic) GK metric?

Symplectic type: Morita equivalence of Hitchin Poisson structures

Special case: $\mathbb{T}_A$ is symplectic $\begin{pmatrix} 0 & -\omega^{-1} \\ \omega & 0 \end{pmatrix}$

$\Rightarrow \mathcal{A}_{\pm} \cong TX_{\pm}$ split $\mathcal{E}_{\pm}$

$\Rightarrow$ one foliation is trivial

$\Rightarrow \sigma_{\pm}$ are related by B-field

Pf: $L_{\sigma_+} = \frac{1}{2i} (L_B - L_A)$ $L_A = \Gamma_{-i\omega}$

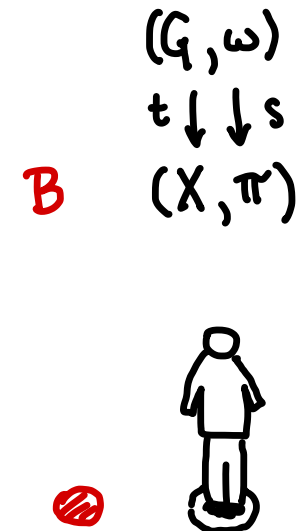
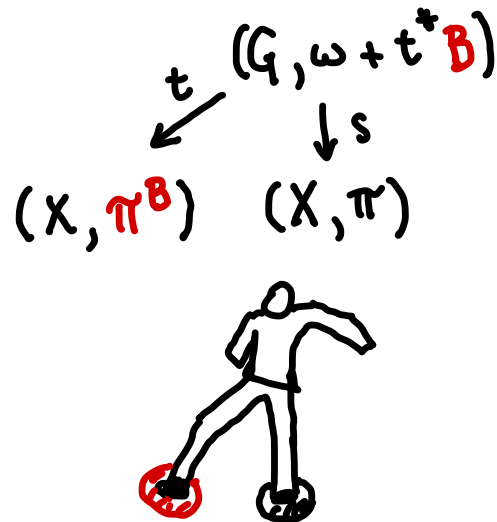
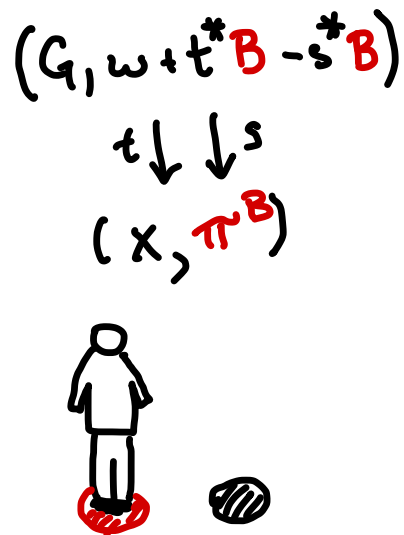
$L_{\sigma_-} = \frac{1}{2i} (L_B - \bar{L}_A)$ $\bar{L}_A = \Gamma_{i\omega}$

$$\Rightarrow L_{\sigma_+} - L_{\sigma_-} = \frac{1}{2i} (\bar{L}_A - L_A)$$

$$= \Gamma_{\omega}$$

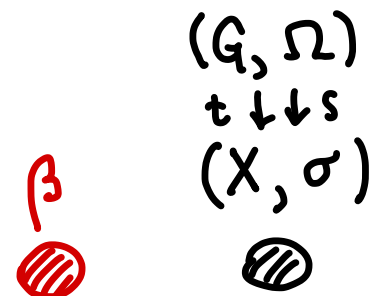
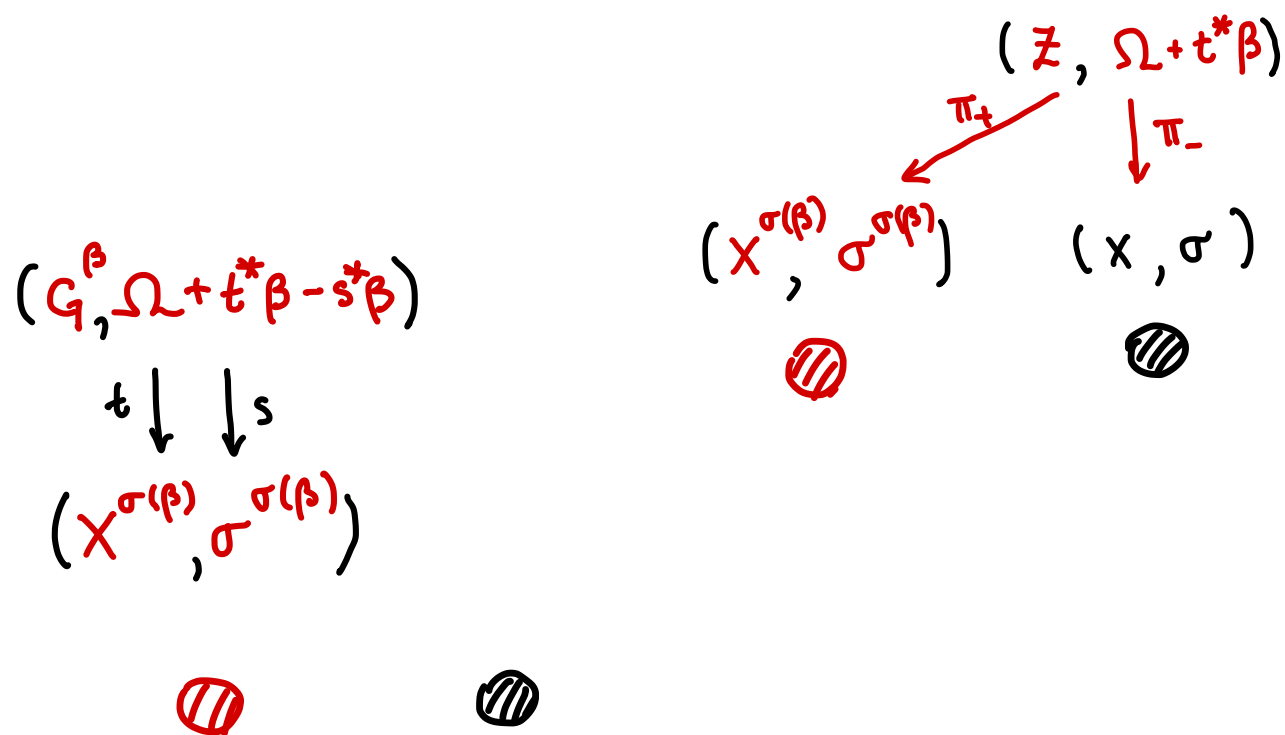
$$\Rightarrow L_{\sigma_+} = e^{\omega} L_{\sigma_-}$$

Main Lemma: the Bursztyn Shuffle



Bursztyn—Radko, Bursztyn—Fernandes

Holomorphic Bursztyn Shuffle



Bailey—G., Bischoff—G.

β : MC element deforming $(G, \Omega) \Rightarrow (X, \sigma)$
 with fixed Morita class.

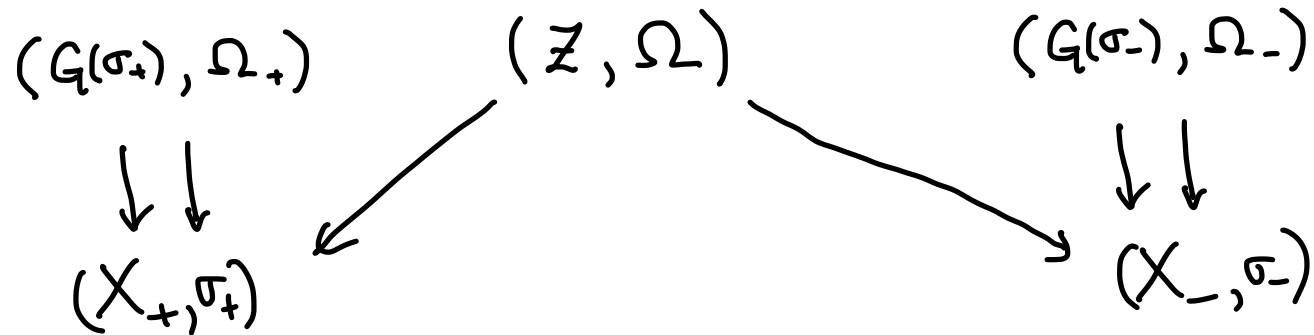
$$\text{MC}(\Omega_x^{\geq 1}, d, [,]_\sigma) \xrightarrow{\sigma} \text{MC}(\mathfrak{X}_x^{\geq 1}, d_\sigma, [,])$$

Ω_x^{12}	$\dots$	$\dots$	$\xrightarrow{(\sigma, \wedge^2 \sigma, \dots)}$	$\mathfrak{X}_x^{12}$	$\dots$	$\dots$
Ω_x^{11}	Ω_x^{21}	$\dots$		$\mathfrak{X}_x^{11}$	$\mathfrak{X}_x^{21}$	$\dots$
Ω_x^{10}	Ω_x^{20}	Ω_x^{30}		$\mathfrak{X}_x^{10}$	$\mathfrak{X}_x^{20}$	$\mathfrak{X}_x^{30}$

Lift to $\text{Def}(G, \Omega)$

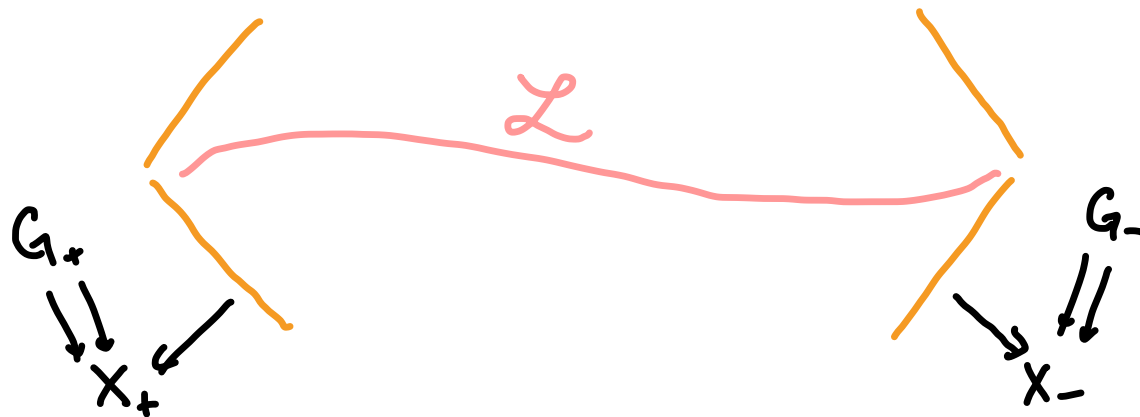
$\text{Def}(X, \sigma)$

Integration in the Symplectic Type Case: (Bischoff—G.—Zabzine)



1. $[(Z, \Omega)] \in \text{Pic}((X_-, \sigma_-), (X_+, \sigma_+))$ is the analog of the Kähler class

2. GK metric: A C^∞ $\text{Im } \Omega$ -Lagrangian bisection



- Single scalar f^n degree of freedom (Kähler potential)

- $\Omega|_L = B\text{-field for shuffle.}$

General case: Integration via decorated surfaces à la Ševera

MODULI SPACES OF FLAT CONNECTIONS AND MORITA EQUIVALENCE OF QUANTUM TORI

PAVOL ŠEVERA

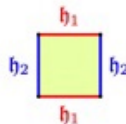
ABSTRACT. We study moduli spaces of flat connections on surfaces with boundary, with boundary conditions given by Lagrangian Lie subalgebras. The resulting symplectic manifolds are closely related with Poisson-Lie groups and their algebraic structure (such as symplectic groupoid structure) gets a geometrical explanation via 3-dimensional cobordisms. We give a formula for the symplectic form in terms of holonomies, based on a central extension of the gauge group by closed 2-forms. This construction is finally used for a certain extension of the Morita equivalence of quantum tori to the world of Poisson-Lie groups.

1. INTRODUCTION

Let $\mathfrak{g}$ be a Lie algebra with an invariant inner product $\langle \cdot, \cdot \rangle$ (of any signature). It gives rise to two interesting types of symplectic manifolds. The first type are moduli spaces of flat $\mathfrak{g}$ -connections on oriented surfaces. The second type are symplectic manifolds connected with Poisson-Lie groups such as the Lu-Weinstein double symplectic groupoid [9] (the symplectic groupoid integrating a Poisson-Lie group) corresponding to a Manin triple

$$\mathfrak{h}_1, \mathfrak{h}_2 \subset \mathfrak{g}.$$

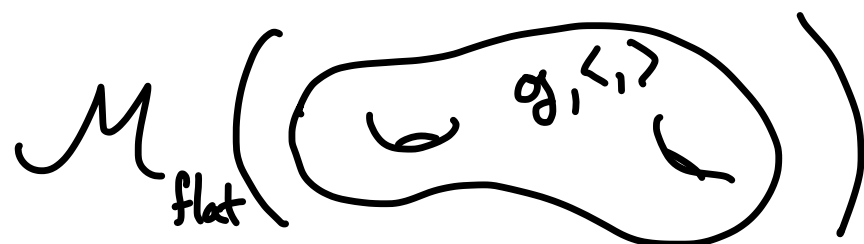
We shall notice that these "Poisson-Lie type" symplectic manifolds are, in fact, themselves moduli spaces of flat connections, if we allow surfaces with boundary and impose boundary conditions on the flat connections. To get the Lu-Weinstein double groupoid, the surface is a square, with boundary conditions as on the picture:



We shall provide formulas for symplectic forms using holonomies of the flat connections, in the spirit of Alekseev-Malkin-Meinrenken [1]. The basic idea is that the symplectic form can be interpreted as the integral over the surface of the curvature of a certain connection. The integral is then readily computed in terms of parallel transport. Moreover we shall describe how 3dim bodies give rise to Lagrangian submanifolds; for example, this picture gives one of the two products in Lu-Weinstein double groupoid:



General case: Integration via decorated surfaces à la Ševera



Symplectic



Poisson



Symplectic

$h_i \subset \text{of Lagrangian}$

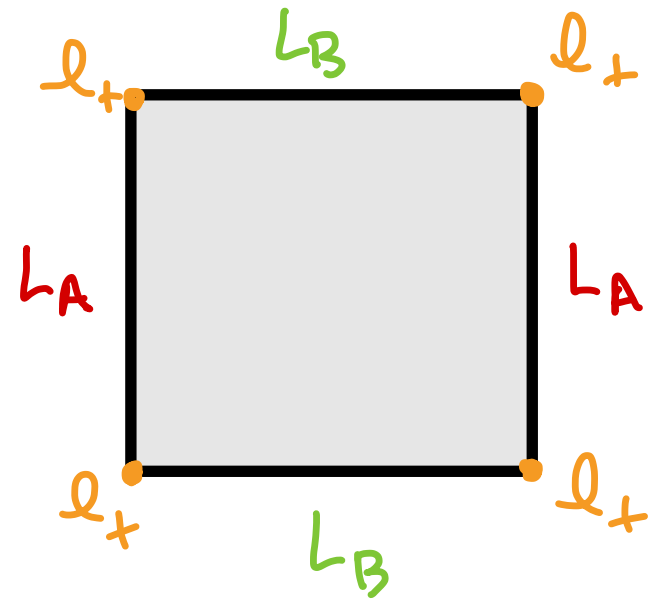
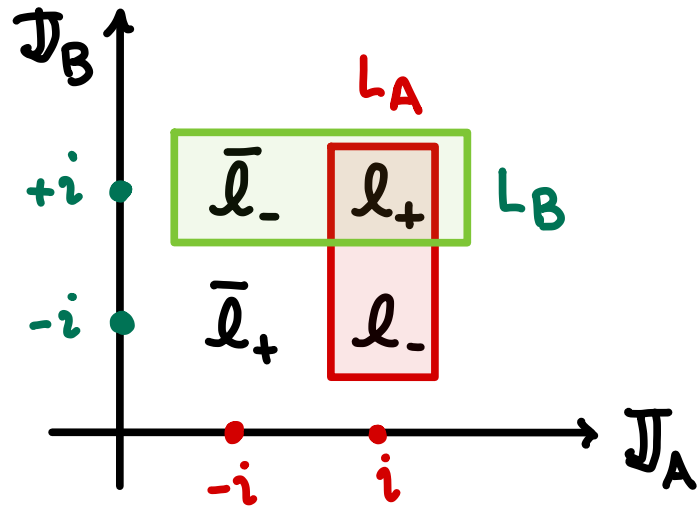
$$h_+ \cap h_- = 0$$

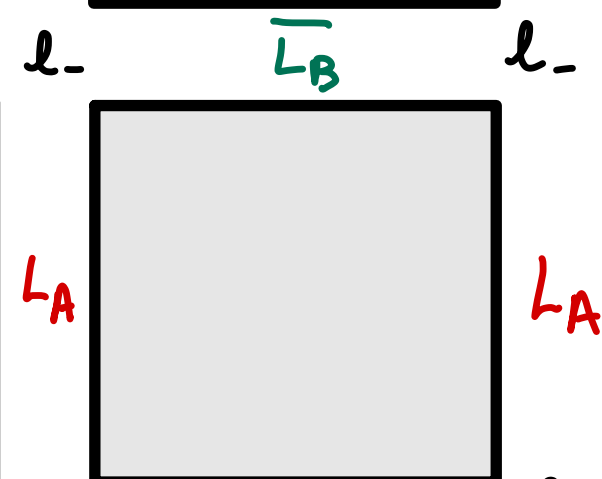
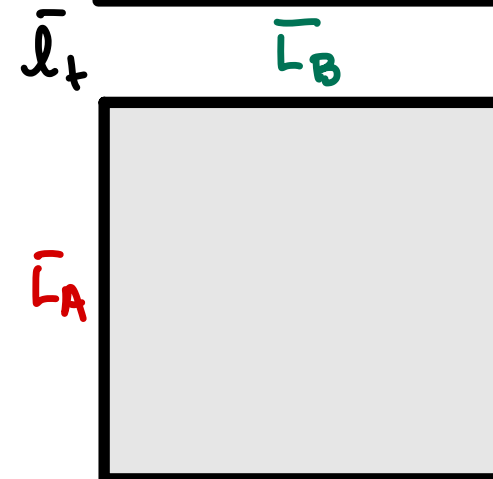
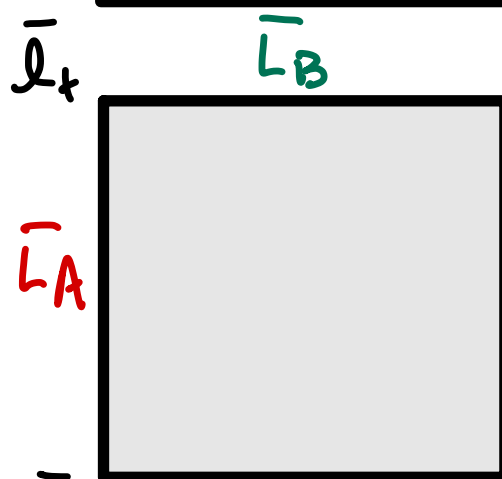
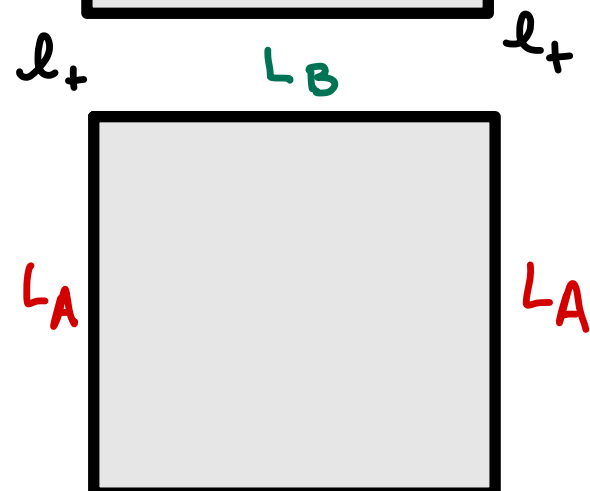
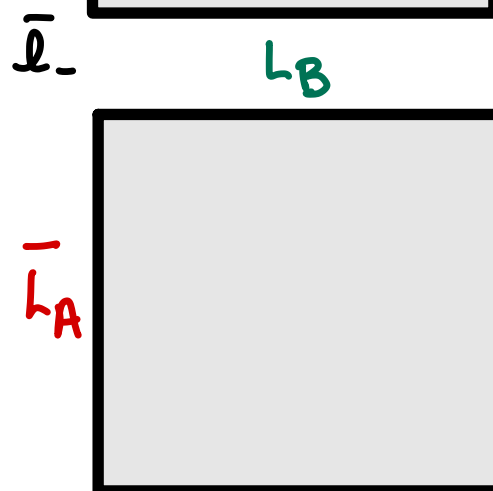
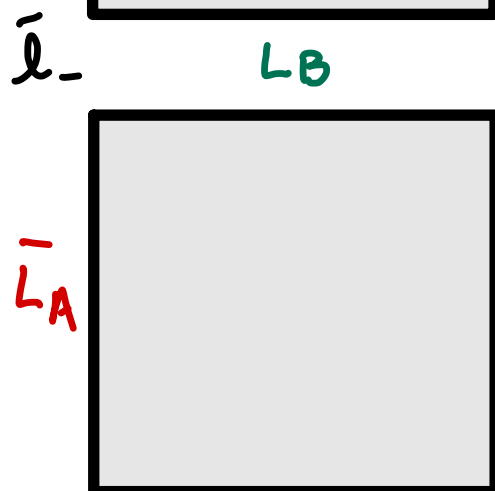
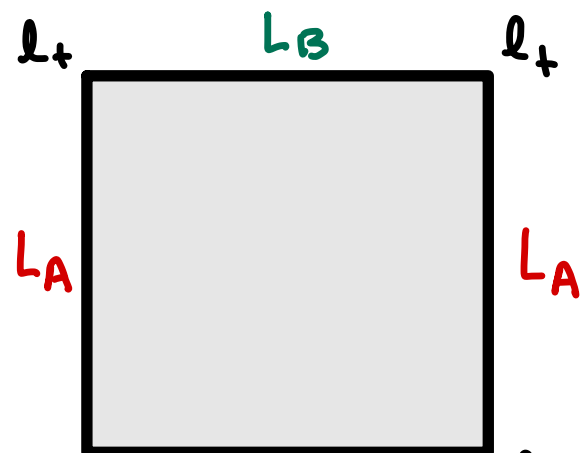
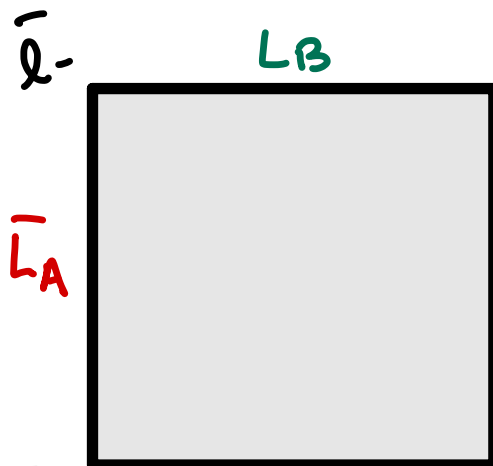
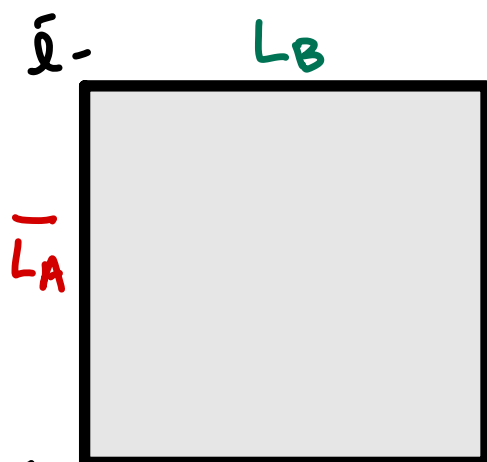
Special Case: Ševera's construction of the Lu—Weinstein Double Symplectic Groupoid:

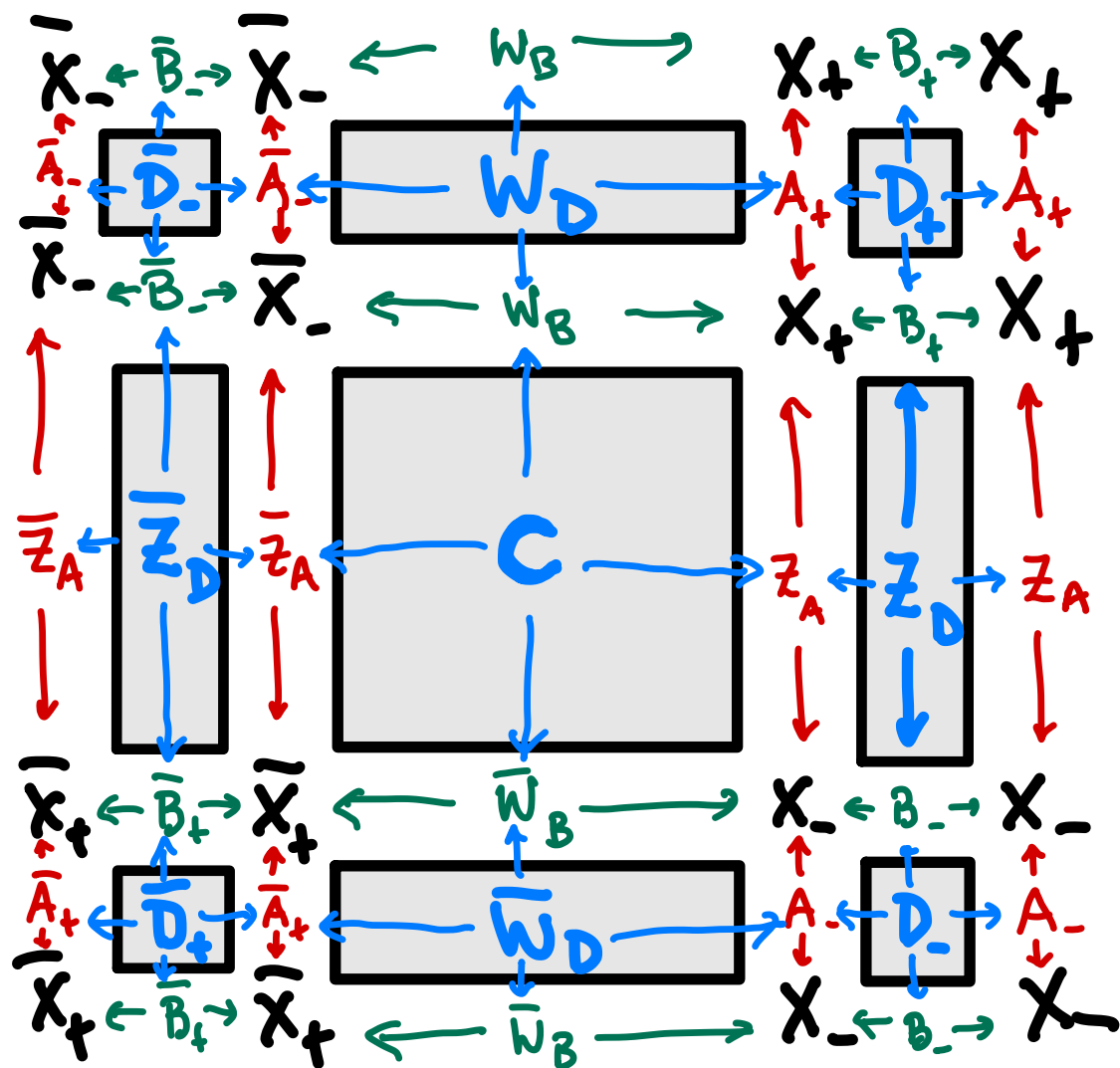
$$(g, \langle, \rangle), \quad g = a \oplus b$$

$$\mathcal{M} \left(\begin{array}{c} \text{a} \\ \text{b} \left[\text{b} \left(\text{g}, \langle, \rangle \right) \text{b} \right] \text{a} \end{array} \right) = \{ (a_1, a_2, b_1, b_2) \in A^2 \times B^2 : a_1 b_1 = b_2 a_2 \}$$

In our case, we have complex Lie algebroids





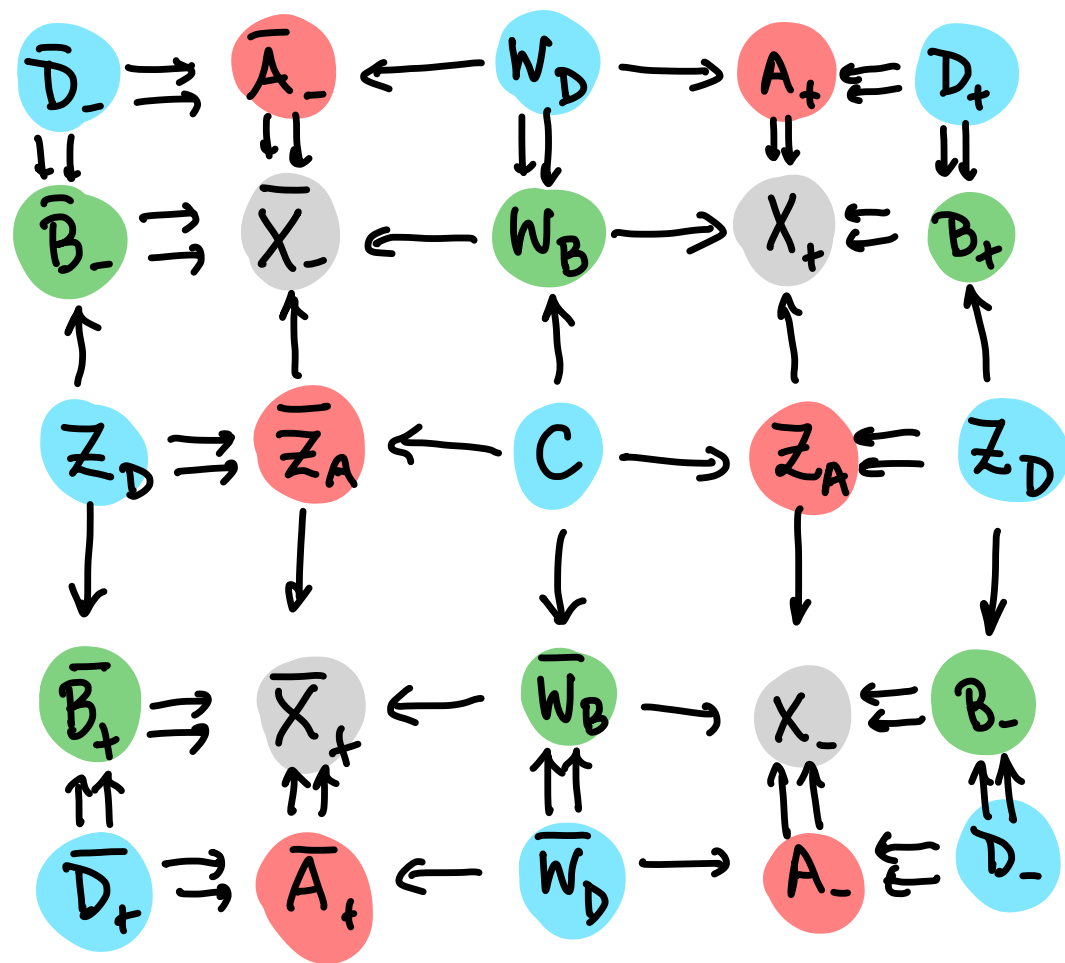


Manin Triple

Horizontal Poisson Groupoids and Morita

Vertical Poisson Groupoids and Morita.

Double Symplectic groupoids and Morita equivalences

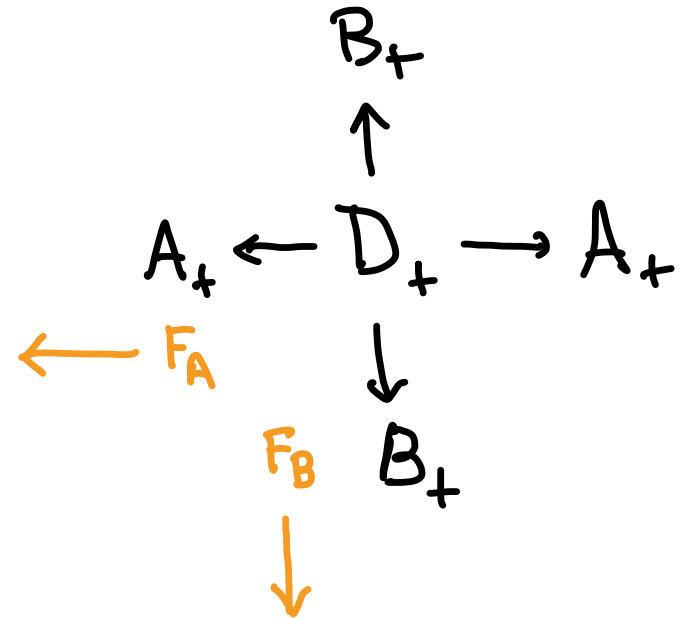


2D Bursztyn Shuffle: A Square Dance

Modify D_+

using $F_A \in \Omega^2(A_+)$

$F_B \in \Omega^2(B_+)$

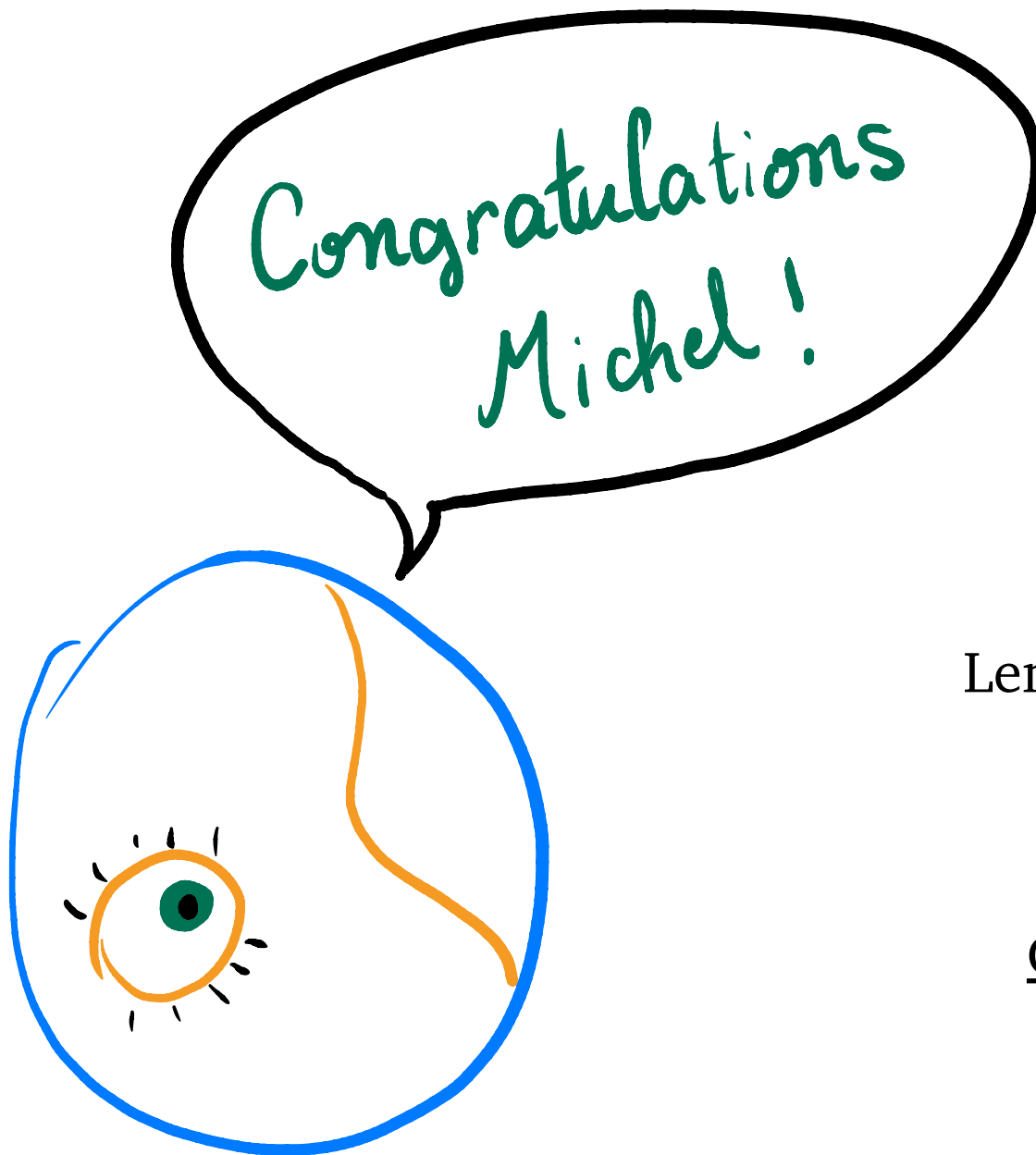


B-fields from MC elts in X_+

⇒ Construct all q of the Double symplectic spaces

⇒ Lagrangian brane bisections determine g





Credits

Current project

Yucong Jiang
Daniel Alvarez

Connected projects

Lennart Döppenschmitt
Francis Bischoff
Maxim Zabzine

Conceptual guidance

Pavol Ševera